DS 420, Project #1

There will be two projects in this course. In general, you will work on **the same dataset** in both projects. You may want to use other datasets for project #2, but be careful as you may not have enough time to work on both datasets. My suggestion is to find a good dataset for both.

Some useful courses

https://www.kaggle.com/learn/pandas

https://www.kaggle.com/learn/data-cleaning

https://www.kaggle.com/learn/feature-engineering

https://www.kaggle.com/learn/python

https://www.kaggle.com/learn/machine-learning-explainability

https://www.kaggle.com/learn/geospatial-analysis

Project Proposal

For this part, submit a pdf file with the requested information below filled in.

- Team members: You need to find your teammate in the class: two or three students in a team.
- Data set to analyze

    - Description/background, how big (#rows, #cols)?, what problem you will work with this dataset (e.g., classification, clustering, regression, …)

    - Link of the dataset:

    - Target – Can be qualitative or quantitative. If qualitative, list the categories

    - Predictor variables (features). Can be a mixture of quantitative, datetime, categorical, and text.

Project #1

The purpose of Project 1 is to implement a machine learning project on a suitable data set that is of interest to your team.

You need to find a dataset yourself. The data set should have at least 20,000 samples. There is no absolute maximum size. The label (response) variable can be either qualitative or quantitative. You will implement and compare ML methods: choosing from linear/logistic regression, decision, support vector machine, …. You should use mini-batch GD to get good model performances and accelerate the training time.

You should include as many of the end-to-end steps (Chapter 2) as possible.

Requirements: **1 submission in a .zip per group.**

1. (20 points) **Exploratory Data Analysis** (01 code file, name it "dataset_EDA", for example "housing_EDA.ipynb"). Your team needs to work together in this step. You do not need to exhaustedly explore/transform the data set in this step. Instead, you need to make data in numbers and scale them. You can come back to this step later when you run ML algorithms on the prepared data and you think you need to improve it. The point is that you should quicky make data ready for algorithms to run on their original version first. Then you may need to make transformations later on. Use assignment01 for your reference.
2. (20 points) **Data preparation** (01 code file, name it "dataset_preparation.py"). Make pipeline(s) to handle as many steps as possible. You will need to include this code file into your code files in #3 and #4.
3. (20 points) **Algorithm investigation** (01 code file/team member, name it "yourlastname_yourfistname_alg1_alg2_investigation.ipynb"). Each of you in the team needs to investigate at least 02 different algorithms for the data set you choose. Two members cannot use identical an algorithm with the same setting. For example, LinearRegression is only counted for 01 algorithm, but can use SVR with RBF kernels and SVR with Polynomial kernels are different, can be counted for 02 algorithms.
4. (20 points) **Train algorithms**. (01 code file / team member, name it "yourlastname_yourfirstname_alg1.ipynb). All members pick top 02/03 algorithms from the previous step and train them. Each member train one of the top algorithms. Here, training algorithms means finding the best hyper parameters for them. Use grid search for hyper tuning. In this step, you may need to come back to #1 (EDA) to refine your data for training. If your group has 02 members, pick top 02 algorithms. If your group has 03 members, pick top 03 algorithms.
5. (20 points) **Conclusion**: all members: make a conclusion file: name it conclusion.ipynb
    a. State clearly the desired performance of the model for the data set. For example, state-of-the art models' performances, human-level performance, … You need to refer to a source for that (e.g., a link). Your models' performances need to be close to the desired performances.
    b. Make comparisons for the algorithms you trained in the previous step (use table or plots for audiences to better see the results). What is your reflection? What metrics do you prioritize to evaluate your models? why?
6. For all steps #1 - #4: Need to outline sub steps, make comments and (or) conclusions clearly to get full credits (deduction: 3 -5 points/task if you miss this requirement)

Here is the timetable

Time	Tasks you need to do	Documents you need to submit	Where to submit	Resources	Notes
Week 4	Project proposal	Project Proposal	Katie, turn in pdf files	See links in chapter 2 of the textbook for datasets. Or any other data sources you can get a dataset from	You should submit your proposals as soon as you can. Once you submit your proposals, I will let you know if they are accepted
End of week 4					You get my approvals on your datasets via Katie

Week 5, 6, 7	Project #1	Code files	Katie, turn in one .zip file	Lectures, assignments, …	
Week 8 -> 14	Project #2	Code files	Katie, turn in one .zip file	Lectures, assignments, …	